

Technology Stack Document

AI Contract & Legal Document Analyzer

1. Project Overview

The AI Contract & Legal Document Analyzer is an AI-powered web application designed to simplify complex legal documents by analyzing contracts, extracting important clauses, identifying potential risks, and generating easy-to-understand summaries.

The project uses modern web technologies with React.js for the frontend, Supabase for backend services, JavaScript-based algorithms for document analysis, and cloud deployment using Vercel.

2. Frontend Technology

React.js

Purpose:

React.js is used to build a dynamic, responsive, and user-friendly interface for the legal document analysis platform.

Usage:

- Building user interface components
- Creating dashboard views
- Managing application state
- Displaying document analysis results
- Creating interactive user experiences

Benefits:

- Component-based architecture
- Faster development
- Reusable UI components
- Efficient rendering

JavaScript (ES6+)

Purpose:

JavaScript is used as the primary programming language for implementing application logic and AI-related processing.

Usage:

- Document processing workflows
- API integration
- Data handling
- Implementing analysis algorithms
- User interaction logic

Benefits:

- Widely supported
 - Works efficiently with React.js
 - Large ecosystem of libraries
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HTML5 & CSS3

Purpose:

Used for creating the structure and styling of the web application.

Usage:

- Page structure
 - Layout design
 - Responsive styling
 - User interface customization
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3. Backend Technology

Supabase

Purpose:

Supabase is used as the backend-as-a-service platform for database management, authentication, and file storage.

Services Used:**Supabase Authentication**

- User registration and login
- Secure user sessions
- Access control

Supabase Database (PostgreSQL)

- Store user information
- Store uploaded document details
- Store analysis results
- Manage application data

Supabase Storage

- Store uploaded legal documents
- Manage document files securely

Benefits:

- Reduces backend development time
 - Provides secure cloud infrastructure
 - Easy integration with React.js
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4. AI & Document Analysis Technologies

JavaScript-Based AI Algorithms

Purpose:

Used to analyze legal documents and extract meaningful information.

Algorithms and Techniques:**Text Processing Algorithms**

- Text cleaning and preprocessing
- Sentence segmentation
- Keyword extraction

Clause Identification

- Pattern matching
- Rule-based detection
- Keyword-based classification

Risk Analysis

- Detection of risky keywords and phrases
- Identification of unusual contract terms
- Risk scoring mechanism

Document Summarization

- Extractive summarization techniques
 - Important sentence identification
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Natural Language Processing (NLP)

Purpose:

NLP techniques are used to understand and process human language present in legal documents.

Applications:

- Understanding legal terminology
 - Extracting important information
 - Identifying relationships between clauses
 - Generating simplified explanations
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5. Document Processing Technologies

PDF/DOCX Processing Libraries

Purpose:

Used for extracting text from uploaded legal documents.

Functions:

- Read uploaded documents
 - Convert documents into machine-readable text
 - Prepare content for analysis
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6. Database Management

PostgreSQL (Through Supabase)

Purpose:

Used for structured data storage.

Stored Data:

- User profiles
 - Document metadata
 - Analysis history
 - Extracted information
 - Risk reports
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7. Version Control & Collaboration

GitHub

Purpose:

Used for managing source code and maintaining project versions.

Usage:

- Code repository management
- Version tracking
- Collaboration
- Project documentation

Benefits:

- Backup of project source code
 - Easy deployment integration
 - Industry-standard workflow
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8. Deployment & Hosting

Vercel

Purpose:

Used for hosting and deploying the React.js application.

Usage:

- Deploy frontend application
- Automatic deployment from GitHub
- Continuous updates after code changes

Benefits:

- Fast deployment
 - Optimized for React applications
 - Simple CI/CD workflow
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9. Development Environment

Visual Studio Code

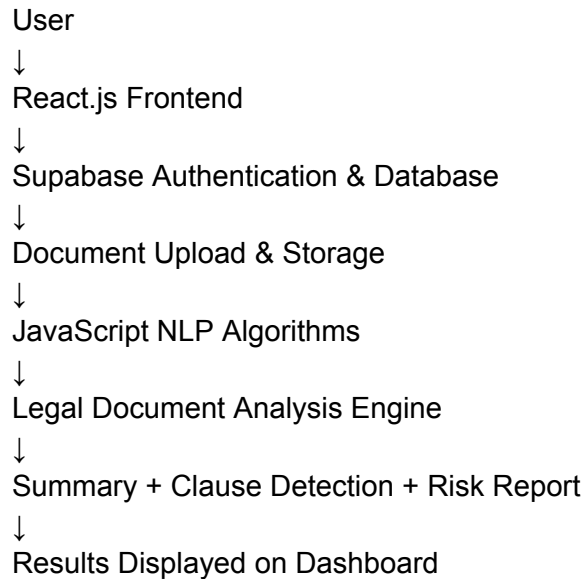
Purpose:

Used as the primary code editor for developing the application.

Usage:

- Writing React components
- Debugging JavaScript code
- Managing project files

10. System Architecture



11. Final Technology Stack Summary

Category	Technology
Frontend	React.js
Programming Language	JavaScript (ES6+)
Styling	HTML5, CSS3
Backend Services	Supabase
Database	PostgreSQL (Supabase)
Authentication	Supabase Auth
File Storage	Supabase Storage
AI Processing	JavaScript NLP Algorithms

Document Processing	PDF/DOCX Parser Libraries
Version Control	GitHub
Deployment	Vercel
Development Tool	VS Code

Conclusion

The selected technology stack provides a complete full-stack development environment for building an AI-powered legal document analysis platform. React.js enables a modern user interface, Supabase provides scalable backend services, JavaScript algorithms handle intelligent document processing, and GitHub with Vercel ensures efficient development and deployment.